

LAPORAN
PRAKTIKUM 1 – SISTEM OPERASI



Dosen pengampu
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1. Operating System Service

Praktik 1.1 menjelajahi layanan OS

```
divo@MSI: ~  
Installing, this may take a few minutes...  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: divo  
New password:  
Retype new password:  
passwd: password updated successfully  
Installation successful!  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
divo@MSI:~$ uname -a  
Linux MSI 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
divo@MSI:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description: Ubuntu 24.04.1 LTS  
Release: 24.04  
Codename: noble  
divo@MSI:~$ ps aux | head -20  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.7  0.1 21700 12812 ?        Ss   22:40   0:00 /sbin/init  
root         2  0.0  0.0   3072 1792 ?        Ss   22:40   0:00 /init  
root         6  0.0  0.0   3088 1792 ?        Ss   22:40   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate  
root       52  0.2  0.2  42248 15616 ?        S<s  22:40   0:00 /usr/lib/systemd/systemd-journald  
root       98  0.2  0.0  25400  6528 ?        Ss   22:40   0:00 /usr/lib/systemd/systemd-udevd  
systemd+ 169  0.1  0.1 21462 12800 ?        Ss   22:40   0:00 /usr/lib/systemd/systemd-resolved  
systemd+ 172  0.1  0.1  91020  7680 ?        Ssl  22:40   0:00 /usr/lib/systemd/systemd-timesyncd  
root      178  0.0  0.0   4236 2560 ?        Ss   22:40   0:00 /usr/sbin/cron -f -P  
message+ 179  0.0  0.0   9528 4992 ?        Ss   22:40   0:00 @dbus-daemon system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only  
root      186  0.1  0.1 17960  8320 ?        Ss   22:40   0:00 /usr/lib/systemd/systemd-logind  
root     193  0.1  0.1 1755840 12032 ?       Ssl  22:40   0:00 /usr/libexec/wsl-pro-service -vv  
syslog    213  0.0  0.0  22588  5248 ?        Ssl  22:40   0:00 /usr/sbin/rsyslogd -n -iNONE  
root     216  0.0  0.0   3116 1792 tty1     Ss+  22:40   0:00 /sbin/agetty -o -p -- \u --noclear - linux  
root     219  0.1  0.3 186996 22328 ?       Ssl  22:40   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal  
root     295  0.0  0.0   3076 1024 ?        Ss   22:40   0:00 /init  
root     296  0.0  0.0   3092 1152 ?        S   22:40   0:00 /init  
divo     297  0.0  0.0   6072  5248 pts/0    Ss   22:40   0:00 -bash  
root     298  0.0  0.0   4224 pts/1    Ss   22:40   0:00 /bin/login -f  
divo     332  0.1  0.1 20312 11008 ?       Ss   22:40   0:00 /usr/lib/systemd/systemd --user  
  
divo@MSI:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:          6 GiB         634Mi       4.6GiB         0Mi         204Mi       4.6GiB  
Swap:         2.0GiB          0B         2.0GiB                   0  
  
divo@MSI:~$ lscpu  
Architecture: x86_64  
CPU op-mode(s): 32-bit, 64-bit  
Address sizes: 48 bits physical, 48 bits virtual  
Byte Order: Little Endian  
CPU(s): 12  
On-line CPU(s) list: 0-11  
Vendor ID: AuthenticAMD  
Model name: AMD Ryzen 5 7535HS with Radeon Graphics
```

Praktik 1.2: Program C - Menggunakan Layanan OS

```
divo@MSI: ~  
acted: BMI Not affected  
Srbds: Not affected  
Tex async abort: Not affected  
divo@MSI:~$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda         8:0    0 388.9M 1 disk  
sdb         8:16   0 160.0 1 disk  
sdc         8:32   0 20.0 0 disk [SWAP]  
sdd         8:40   0 17.0 0 disk  
sde         8:64   0 17.0 0 disk /mnt/eslg/distro  
  
divo@MSI:~$ nano os_services.c  
divo@MSI:~$ gcc -o os_services os_services.c  
Command gcc not found, but can be installed with:  
sudo apt install gcc  
divo@MSI:~$ sudo apt install gcc  
[sudo] password for divo:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc-13  
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libasan8  
  libatomic1 libitm1 libubsan1 libgcc-13-dev libgcc-s1 libgomp1  
  libhwasan1 libisl23 liblibm1 libquadmath0 libstdc++6  
  libubsan1 libubsan1 libubsan1 libubsan1 libubsan1 libubsan1  
Suggested packages:  
  cpp-doc gcc-13-locales cpp-13-doc gcc-multilib make autoconf automake  
  libtool flex bison g++-13 g++-13-x86-64-linux-gnu g++-13  
  g++-13-x86-64-linux-gnu libgcc-13-dev libgcc-13-dev libgcc-13-dev  
  libhwasan1 libhwasan1 libhwasan1 libhwasan1 libhwasan1 libhwasan1  
  libhwasan1 libhwasan1 libhwasan1 libhwasan1 libhwasan1 libhwasan1  
The following NEW packages will be installed:  
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc-13  
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libasan8  
  libatomic1 libitm1 libubsan1 libgcc-13-dev libgcc-s1 libgomp1  
  libhwasan1 libisl23 liblibm1 libquadmath0 libstdc++6  
  libubsan1 libubsan1 libubsan1 libubsan1 libubsan1 libubsan1  
Need to get 55.0 MB of archives.  
After this operation, 100 MB of additional disk space will be used.  
Do you want to continue? [Y/n] Y  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-base amd64 13.3.0-ubuntu2-24.04 [11.5 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libisl23 amd64 0.26-1build1 [680 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libgcc1 amd64 13.3.0-10ubuntu1 [45.5 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-x86-64-linux-gnu amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13-x86-64-linux-gnu amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc1 amd64 13.3.0-ubuntu2-24.04 [45.5 kB]  
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libasan8 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libatomic1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libitm1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libquadmath0 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++6 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 13.3.0-ubuntu2-24.04 [10.7 MB]  
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-x86-64-linux-gnu amd64 13.3.0-ubuntu2-24.04 [10.7 MB]
```

Compile dan jalankan

```
divo@MSI: ~$
Generating locales (this might take a while)...
Generation complete.
Setting up rpcsvc-proto (1.4.2-ubuntu7) ...
Setting up gcc-13-base:amd64 (13.3.0-ubuntu2-24.04) ...
Setting up libquadmath0:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libncurses6:amd64 (6.3-1build1) ...
Setting up libatomic1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libasan1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libbssan1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libcrypt-dev:amd64 (4.4.36-4build1) ...
Setting up libsan1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libtana2:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libl12:amd64 (8.26-1build1) ...
Setting up libstdc++6:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libc-dev-bin (2.39-ubuntu6) ...
Setting up libl1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libl1a:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libl1b:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up cpp-13-x86_64-linux-gnu (13.3.0-ubuntu2-24.04) ...
Setting up libgcc-13-dev:amd64 (13.3.0-ubuntu2-24.04) ...
Setting up libc-dev:amd64 (2.39-ubuntu6) ...
Setting up cpp-x86_64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up gcc-13-x86_64-linux-gnu (13.3.0-ubuntu2-24.04) ...
Setting up gcc-13 (13.3.0-ubuntu2-24.04) ...
Setting up cpp (4:13.2.0-7ubuntu1) ...
Setting up gcc-x86_64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up gcc (4:13.2.0-7ubuntu1) ...
Setting up libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libheif1:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libl1c:amd64 (2.39-ubuntu6) ...
Setting up libheif-plugin-lindec:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.1) ...
Processing triggers for libc-bin (2.39-ubuntu6) ...
Processing triggers for man-db (2.12.0-4build2) ...
divo@MSI:~$ gcc -o os_services os_services.c
divo@MSI:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 1983
Parent Process ID (PPID): 297

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 Bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
divo@MSI:~$ pmd
divo@MSI:~$ pmd
pmd: command not found
divo@MSI:~$ cd
divo@MSI:~$ ls -la
total 56
drwxr-xr-x 4 divo divo 4096 Oct 15 22:52
drwxr-xr-x 3 root root 4096 Oct 15 22:48
-rw-r--r-- 1 divo divo 228 Oct 15 22:48 bash_logout
-rw-r--r-- 1 divo divo 228 Oct 15 22:48 bash_logout
-rw-r--r-- 1 divo divo 3771 Oct 15 22:48 bashrc
```

2. User and Operating-System Interface

Praktik 2.1: Command Line Interface (CLI)

```
divo@MSI: ~$
=== Program Execution Service ===
Process ID (PID): 1983
Parent Process ID (PPID): 297

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 Bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
divo@MSI:~$ pmd
divo@MSI:~$ pmd
pmd: command not found
divo@MSI:~$ cd
divo@MSI:~$ cd /tmp
divo@MSI:~/tmp$ mkdir test_dir
divo@MSI:~/tmp$ cd test_dir
divo@MSI:~/tmp/test_dir$ ls -l | grep "d"
d
divo@MSI:~/tmp/test_dir$ echo "Hello World" > output.txt
divo@MSI:~/tmp/test_dir$ cat output.txt
Hello World
divo@MSI:~/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
divo@MSI:~/tmp/test_dir$ cat < output.txt
Hello World
divo@MSI:~/tmp/test_dir$ sleep 30 &
[1] 2186
divo@MSI:~/tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
divo@MSI:~/tmp/test_dir$ fg %1
sleep 30
divo@MSI:~/tmp/test_dir$ nano simple_shell.c
divo@MSI:~/tmp/test_dir$ gcc -o simple_shell simple_shell.c
divo@MSI:~/tmp/test_dir$ ./simple_shell
myshe11 > C
divo@MSI:~/tmp/test_dir$ cd
divo@MSI:~$ nano process_syscalls.c
divo@MSI:~$ ls
os_services  os_services.c  process_syscalls.c  test_file.txt
divo@MSI:~$ nano file_syscalls.c
divo@MSI:~$ ls
file_syscalls.c  os_services.c  test_file.txt
```

Praktik 2.2: Program C - Shell Sederhana

```
divo@MSI: -
[1]+  Running                  sleep 30 &
divo@MSI:/tmp/test_dir$ fg %1
sleep 30
divo@MSI:/tmp/test_dir$ nano simple_shell.c
divo@MSI:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
divo@MSI:/tmp/test_dir$ ./simple_shell
myshell> ^C
divo@MSI:/tmp/test_dir$ cd
divo@MSI:~$ nano process_syscalls.c
divo@MSI:~$ ls
os_services  os_services.c  process_syscalls.c  test_file.txt
divo@MSI:~$ nano file_syscalls.c
divo@MSI:~$ ls
file_syscalls.c  os_services.c  test_file.txt
divo@MSI:~$ nano info_syscalls.c
divo@MSI:~$ nano info_syscalls.c
divo@MSI:~$ gcc -o process_syscalls process_syscalls.c
divo@MSI:~$ gcc -o file_syscalls file_syscalls.c
divo@MSI:~$ gcc -o info_syscalls info_syscalls.c
divo@MSI:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 2233
Parent waiting for child (PID: 2234)...
Child PID: 2234
Child's Parent PID: 2233
Child executing 'ls' command...
total 88
-rwxr-xr-x 1 divo divo 16352 Oct 15 23:18 file_syscalls
-rw-r--r-- 1 divo divo   966 Oct 15 23:15 file_syscalls.c
-rwxr-xr-x 1 divo divo 16384 Oct 15 23:18 info_syscalls
-rw-r--r-- 1 divo divo   712 Oct 15 23:17 info_syscalls.c
-rwxr-xr-x 1 divo divo 16488 Oct 15 22:51 os_services
-rw-r--r-- 1 divo divo  1315 Oct 15 22:48 os_services.c
-rwxr-xr-x 1 divo divo 16360 Oct 15 23:17 process_syscalls
-rw-r--r-- 1 divo divo   831 Oct 15 23:13 process_syscalls.c
-rw-r--r-- 1 divo divo    23 Oct 15 22:52 test_file.txt
Child exited with status: 0
divo@MSI:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
```

3. System Calls Praktik

3.1: Process Control System Calls

Buat file process_syscalls.c:

Praktik 3.2: File Management System Calls

Buat file file_syscalls.c:

Praktik 3.3: Information Maintenance System Calls

Buat file info_syscalls.c:

Compile smua:

```
divo@MSI: ~$ cat output.txt
Hello world
divo@MSI: /tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
divo@MSI: /tmp/test_dir$ cat < output.txt
Hello world
divo@MSI: /tmp/test_dir$ sleep 30 &
[1] 2188
divo@MSI: /tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
divo@MSI: /tmp/test_dir$ fg %1
sleep 30
divo@MSI: /tmp/test_dir$ nano simple_shell.c
divo@MSI:~$ gcc -o process_syscalls process_syscalls.c test_file.txt
divo@MSI:~$ nano file_syscalls.c
divo@MSI:~$ ls
os_services.c  process_syscalls.c  test_file.txt
divo@MSI:~$ ls
file_syscalls.c  os_services.c      process_syscalls.c
divo@MSI:~$ nano info_syscalls.c
divo@MSI:~$ gcc -o process_syscalls process_syscalls.c
divo@MSI:~$ gcc -o file_syscalls file_syscalls.c
divo@MSI:~$ gcc -o info_syscalls info_syscalls.c
divo@MSI:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 2233
Parent waiting for child (PID: 2234)....
Child PID: 2234
Child's Parent PID: 2233
Child executing 'ls' command...
Total 88
-rwxr-xr-x 1 divo divo 16392 Oct 15 22:18 file_syscalls
-rwxr-xr-x 1 divo divo 960 Oct 15 22:15 file_syscalls.c
-rwxr-xr-x 1 divo divo 16384 Oct 15 22:18 info_syscalls
-rwxr-xr-x 1 divo divo 712 Oct 15 22:17 info_syscalls.c
-rwxr-xr-x 1 divo divo 16488 Oct 15 22:11 os_services
-rwxr-xr-x 1 divo divo 1318 Oct 15 22:48 os_services.c
-rwxr-xr-x 1 divo divo 16368 Oct 15 22:15 process_syscalls
-rwxr-xr-x 1 divo divo 831 Oct 15 22:13 process_syscalls.c
-rwxr-xr-x 1 divo divo 21 Oct 15 22:52 test_file.txt
Child exited with status: 0
divo@MSI:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
divo@MSI:~$ ./info_syscalls
=== Information Maintenance System Calls ===
Process ID: 2248
Parent Process ID: 297
User ID: 1000
Group ID: 1000
Current time: Wed Oct 15 22:18:05 2025
Clock ticks since boot: 1718495247
Sleeping for 2 seconds...
Anshu!
divo@MSI:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
UNIT
```

4. System Services Praktik

4.1: Menggunakan systemd

```
divo@MSI: ~$ systemctl list-units --type=service
Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of
SUB → The low-level unit activation state, values depend on unit type

30 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
divo@MSI:~$ sudo systemctl reset-failed
divo@MSI:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-setup.service              loaded active exited Set console
cron.service                       loaded active running Regular ba
dbus.service                       loaded active running D-Bus Syst
getty@tty1.service                loaded active running Getty on t
keyboard-setup.service             loaded active exited Set the co
kmod-static-nodes.service          loaded active exited Create Lis
polkit.service                    loaded active running Authorizat
rsyslog.service                   loaded active running System Log
setvtrgb.service                  loaded active exited Set consol
snapd.seeded.service              loaded active exited Wait until
systemd-journal-flush.service      loaded active exited Flush Jour
systemd-journald.service           loaded active running Journal Se
systemd-logind.service             loaded active running User Login
systemd-modules-load.service       loaded active exited Load Kerne
systemd-remount-fs.service         loaded active exited Remount Ro
systemd-resolved.service           loaded active running Network Na
systemd-sysctl.service             loaded active exited Apply Kerna
systemd-timesyncd.service          loaded active running Network Ti
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Sta
systemd-tmpfiles-setup-dev.service loaded active exited Create Sta
systemd-tmpfiles-setup.service     loaded active exited Create Vol
systemd-udev-trigger.service       loaded active exited Coldplug A
systemd-udevd.service              loaded active running Rule-based
systemd-update-utmp.service        loaded active exited Record Sys
systemd-user-sessions.service      loaded active exited Permit Use
unattended-upgrades.service        loaded active running Unattended
user-runtime-dir@1000.service      loaded active exited User Runti
user@1000.service                 loaded active running User Manag
wsl-pro.service                   loaded active running Bridge to

Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of
SUB → The low-level unit activation state, values depend on unit type

29 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
divo@MSI:~$ systemctl status ssh
```

```
divo@MSI:~$ journalctl -u ssh -n 20
Oct 15 23:30:47 MSI systemd[1]: Starting ssh.service - OpenBSD Secure Shell>
Oct 15 23:30:47 MSI sshd[3200]: Server listening on 0.0.0.0 port 22.
Oct 15 23:30:47 MSI sshd[3200]: Server listening on :: port 22.
Oct 15 23:30:47 MSI systemd[1]: Started ssh.service - OpenBSD Secure Shell >
divo@MSI:~$ journalctl -u ssh -n 20
Oct 15 23:30:47 MSI systemd[1]: Starting ssh.service - OpenBSD Secure Shell>
Oct 15 23:30:47 MSI sshd[3200]: Server listening on 0.0.0.0 port 22.
Oct 15 23:30:47 MSI sshd[3200]: Server listening on :: port 22.
Oct 15 23:30:47 MSI systemd[1]: Started ssh.service - OpenBSD Secure Shell >
lines 1-4/4 (END)
divo@MSI:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    PRESET
apparmor.service                        enabled enabled
apport.service                         enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                    enabled enabled
cloud-init-local.service                enabled enabled
cloud-init.service                     enabled enabled
console-setup.service                  enabled enabled
cron.service                           enabled enabled
dmesg.service                           enabled enabled
e2scrub_reap.service                   enabled enabled
getty@.service                         enabled enabled
keyboard-setup.service                 enabled enabled
landscape-client.service                enabled enabled
networkd-dispatcher.service            enabled enabled
rsyslog.service                        enabled enabled
setvtrgb.service                       enabled enabled
snapd.apparmor.service                 enabled enabled
snapd.autoupdate.service               enabled enabled
snapd.core-fixup.service               enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service                  enabled enabled
snapd.service                          enabled enabled
snapd.systemd-shutdown.service         enabled enabled
ssh.service                            enabled enabled
systemd-pstore.service                 enabled enabled
systemd-resolved.service               enabled enabled
systemd-timesyncd.service              enabled enabled
ub-reboot-cmds.service                 enabled enabled
ubuntu-advantage.service               enabled enabled
unattended-upgrades.service            enabled enabled
wsl-pro.service                        enabled enabled
31 unit files listed.
divo@MSI:~$ nano my_daemon.c
```

Praktik 4.2: Membuat Custom System Service

Buat program sederhana my_daemon.c:

Compile:

Buat service file /etc/systemd/system/my_daemon.service:

Kelola service:

```
divo@MSI:~$ gcc /home/divo/my_daemon.c -o /home/divo/my_daemon
divo@MSI:~$ chmod +x /home/divo/my_daemon
divo@MSI:~$ nano /home/divo/my_daemon.c
divo@MSI:~$ gcc -o my_daemon my_daemon.c
divo@MSI:~$ sudo nano /etc/systemd/system/my_daemon.service
divo@MSI:~$ sudo systemctl daemon-reload
divo@MSI:~$ sudo systemctl start my_daemon
divo@MSI:~$ sudo systemctl enable my_daemon
divo@MSI:~$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-16 00:13:43 WIB; 6s ago
     Main PID: 4618 (my_daemon.sh)
        Tasks: 2 (limit: 8128)
      Memory: 684.0K (peak: 1.2M)
         CPU: 9ms
       CGroup: /system.slice/my_daemon.service
               └─4618 /bin/bash /home/divo/my_daemon.sh
                  4659 sleep 5

Oct 16 00:13:43 MSI systemd[1]: Started my_daemon.service - My Custom Daemon.
divo@MSI:~$ sudo journalctl -u my_daemon
Oct 15 23:41:39 MSI systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 23:41:39 MSI systemd[1]: my_daemon.service: Control process exited, code=exited, stb
Oct 15 23:41:39 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:39 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
Oct 15 23:41:40 MSI systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Control process exited, code=exited, stb
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 15 23:41:40 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
Oct 15 23:41:40 MSI systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Control process exited, code=exited, stb
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 15 23:41:40 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
Oct 15 23:41:40 MSI systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Control process exited, code=exited, stb
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 15 23:41:40 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
Oct 15 23:41:40 MSI systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Control process exited, code=exited, stb
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 15 23:41:40 MSI systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 23:41:40 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
Oct 15 23:41:41 MSI systemd[1]: my_daemon.service: Scheduled restart job, restart counter >
```

5. Linkers and Loaders Praktik

5.1: Proses Linking

Buat tiga file: math_ops.h:

math_ops.c

```
divo@MSI:~$ nano math_ops.h
divo@MSI:~$ nano math_ops.c
divo@MSI:~$ nano main.c
divo@MSI:~$ gcc -E main.c -o main.i
main.c:2:10: fatal error: math_ops.h: No such file or directory
  2 | #include "math_ops.h"
    |
compilation terminated.
divo@MSI:~$ ls
file_syscalls  main.c      my_daemon.c  os_services.c  test_file.txt
file_syscalls.c  math_ops.c  my_daemon.log  process_syscalls
info_syscalls.c  math_ops.y  my_daemon.sh  process_syscalls.c
divo@MSI:~$ nano math_ops.h
divo@MSI:~$ ls
file_syscalls  main.c      my_daemon  os_services  syscall_test.txt
file_syscalls.c  math_ops.c  my_daemon.c  os_services.c  test_file.txt
info_syscalls.c  math_ops.y  my_daemon.sh  process_syscalls.c
divo@MSI:~$ gcc -E main.c -o main.i
divo@MSI:~$ gcc -E math_ops.c -o math_ops.i
divo@MSI:~$ gcc -S main.c -o main.s
divo@MSI:~$ gcc -S math_ops.c -o math_ops.s
divo@MSI:~$ gcc -c main.c -o main.o
divo@MSI:~$ gcc -c math_ops.c -o math_ops.o
divo@MSI:~$ gcc main.o math_ops.o -o program
divo@MSI:~$ nm main.o
0000000000000000 T main
0000000000000000 U add
0000000000000000 U multiply
0000000000000000 U printf
divo@MSI:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
divo@MSI:~$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
0000000000002148 w _ITM_deregisterTMCloneTable
0000000000002148 w _ITM_registerTMCloneTable
000000000000201c r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
00000000000038c r __abi_tag
0000000000004010 B __bss_start
0000000000004000 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start

-rwxr-xr-x 1 divo divo 16K Oct 16 00:46 program_dynamic
-rwxr-xr-x 1 divo divo 767K Oct 16 00:46 program_static
divo@MSI:~$ ldd program_static
not a dynamic executable
divo@MSI:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffe6f576000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007550cc000000)
/lib64/ld-linux-x86-64.so.2 (0x00007550cc000000)
divo@MSI:~$ gcc -c -fPIC math_ops.c -o math_ops.o
divo@MSI:~$ gcc -shared -o libmathops.so math_ops.o
divo@MSI:~$ gcc main.c -L. -lmathops -o program_shared
divo@MSI:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
divo@MSI:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
Tag              Type              Name/Value
0x0000000000000001 (NEEDED)          Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)          Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x1000
0x000000000000000d (FINI)            0x11e4
0x0000000000000019 (INIT_ARRAY)       0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)       0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000006ffffff5 (GNU_HASH)         0x3b0
0x0000000000000005 (STRTAB)           0x4b0
0x0000000000000006 (SYMTAB)           0x3d8
0x000000000000000a (STRSZ)            170 (bytes)
0x000000000000000b (SYMENT)           24 (bytes)
0x0000000000000015 (DEBUG)            0x0
0x0000000000000003 (PLTGOT)           0x3fa8
0x0000000000000002 (PLTRELSZ)         72 (bytes)
0x0000000000000014 (PLTREL)           RELA
0x0000000000000017 (JMPREL)           0x600
0x0000000000000007 (RELA)             0x5a0
0x0000000000000008 (RELA_SZ)          192 (bytes)
0x0000000000000009 (RELAENT)          24 (bytes)
0x000000000000001e (FLAGS)            BIND_NOW
0x000000006ffffffb (FLAGS_1)          Flags: NOW PIE
0x000000006ffffffe (VERNEED)          0x570
0x000000006fffffff (VERNEEDNUM)      1
0x000000006ffffff0 (VERSYM)          0x55a
0x000000006ffffff9 (RELACOUNT)        3
0x0000000000000000 (NULL)            0x0
divo@MSI:~$ nano os_specific.c
divo@MSI:~$ file program
```

Praktik 5.2: Static vs Dynamic Linking

Praktik 5.3: Membuat dan Menggunakan Shared Library

```
divo@MSI:~$ ldd program_static
not a dynamic executable
divo@MSI:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffe6f576000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000075500cc00000)
/lib64/ld-linux-x86-64.so.2 (0x000075500ced5000)
divo@MSI:~$ gcc -c -fPIC math_ops.c -o math_ops.o
divo@MSI:~$ gcc -shared -o libmathops.so math_ops.o
divo@MSI:~$ gcc main.c -L. -lmathops -o program_shared
divo@MSI:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
divo@MSI:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
   Tag              Type              Name/Value
0x0000000000000001 (NEEDED)          Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)          Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x1000
0x000000000000000d (FINI)            0x11e4
0x0000000000000019 (INIT_ARRAY)        0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)        0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000006ffffaf5 (GNU_HASH)         0x3b0
0x0000000000000005 (STRTAB)           0x4b0
0x0000000000000006 (SYMTAB)           0x3d8
0x000000000000000a (STRSZ)            170 (bytes)
0x000000000000000b (SYMENT)           24 (bytes)
0x0000000000000015 (DEBUG)            0x0
0x0000000000000003 (PLTGOT)           0x3fa8
0x0000000000000002 (PLTRELSZ)         72 (bytes)
0x0000000000000014 (PLTREL)           RELA
0x0000000000000017 (JMPREL)           0x60
0x0000000000000007 (RELA)             0x5a0
0x0000000000000008 (RELASZ)           192 (bytes)
0x0000000000000009 (RELAENT)          24 (bytes)
0x000000000000001e (FLAGS)            BIND_NOW
0x000000006fffffb (FLAGS_1)          Flags: NOW PIE
0x000000006fffffe (VERNEED)           0x570
0x000000006fffff (VERNEEDNUM)         1
0x000000006fffff9 (VERSYM)            0x55a
0x000000006fffff9 (RELACOUNT)         3
0x0000000000000000 (NULL)             0x0
divo@MSI:~$ nano os_specific.c
divo@MSI:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc342f23076fc2, for GNU/Linux 3.2.0, not stripped
```


6 .Why Applications Are Operating-System Specific Praktik

6.1: System Call Differences Buat file os_specific.c

Buat file os_specific.c:

Praktik 6.2: Binary Format Analysis

```
divo@MSt:~$ nano os_specific.c
divo@MSt:~$ file program
program: ELF 64-bit LSB pie executable, x86_64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc3
42f23876fc2, for GNU/Linux 3.2.0, not stripped
divo@MSt:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:           ELF64
  Data:            2's complement, little endian
  Version:         1 (current)
  OS/ABI:          UNIX - System V
  ABI Version:     0
  Type:            DYN (Position-Independent Executable file)
  Machine:         Advanced Micro Devices X86-64
  Version:         0x1
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14072 (bytes into file)
  Flags:           0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
divo@MSt:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:

Section Headers:
 [Nr] Name              Type              Address            Offset
     Size              EntSize          Flags              Link Info Align
 [ 0] 0000000000000000 NULL              0000000000000000 0 0 0
 [ 1] .interp             PROGBITS          0000000000000318 00000318
     000000000000001c 0000000000000000 A 0 0 1
 [ 2] .note.gnu.pr[...] NOTE              0000000000000338 00000338
     0000000000000030 0000000000000000 A 0 0 8
 [ 3] .note.gnu.bu[...] NOTE              0000000000000368 00000368
     0000000000000024 0000000000000000 A 0 0 4
 [ 4] .note.ABI-tag       NOTE              000000000000038c 0000038c
     0000000000000020 0000000000000000 A 0 0 4
 [ 5] .gnu.hash            GNU_HASH          00000000000003b0 000003b0
     0000000000000024 0000000000000000 A 0 0 8
 [ 6] .dynsym              DYNSYM            00000000000003d8 000003d8
     00000000000000a8 0000000000000018 A 7 1 8
 [ 7] .dynstr              STRTAB            0000000000000400 00000400
     000000000000008f 0000000000000000 A 0 0 1
```

```
divo@MSt:~$ readelf -S program
Section header string table index: 30
divo@MSt:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:

Section Headers:
 [Nr] Name              Type              Address            Offset
     Size              EntSize          Flags              Link Info Align
 [ 0] 0000000000000000 NULL              0000000000000000 0 0 0
 [ 1] .interp             PROGBITS          0000000000000318 00000318
     000000000000001c 0000000000000000 A 0 0 1
 [ 2] .note.gnu.pr[...] NOTE              0000000000000338 00000338
     0000000000000030 0000000000000000 A 0 0 8
 [ 3] .note.gnu.bu[...] NOTE              0000000000000368 00000368
     0000000000000024 0000000000000000 A 0 0 4
 [ 4] .note.ABI-tag       NOTE              000000000000038c 0000038c
     0000000000000020 0000000000000000 A 0 0 4
 [ 5] .gnu.hash            GNU_HASH          00000000000003b0 000003b0
     0000000000000024 0000000000000000 A 0 0 8
 [ 6] .dynsym              DYNSYM            00000000000003d8 000003d8
     00000000000000a8 0000000000000018 A 7 1 8
 [ 7] .dynstr              STRTAB            0000000000000400 00000400
     000000000000008f 0000000000000000 A 0 0 1
 [ 8] .gnu.version         VERSYM            0000000000000510 00000510
     000000000000000e 0000000000000002 A 6 0 2
 [ 9] .gnu.version_r       VERNEED           0000000000000520 00000520
     0000000000000030 0000000000000000 A 7 1 8
 [10] .rela.dyn            RELA              0000000000000550 00000550
     00000000000000c0 0000000000000018 A 6 0 8
 [11] .rela.plt            RELA              0000000000000610 00000610
     0000000000000018 0000000000000018 A 5 24 8
 [12] .init                PROGBITS          0000000000001000 00001000
     000000000000001b 0000000000000000 AX 0 0 4
 [13] .plt                 PROGBITS          0000000000001020 00001020
     0000000000000020 0000000000000010 AX 0 0 16
 [14] .plt.got             PROGBITS          0000000000001040 00001040
     0000000000000010 0000000000000010 AX 0 0 16
 [15] .plt.sec             PROGBITS          0000000000001050 00001050
     0000000000000010 0000000000000010 AX 0 0 16
 [16] .text                PROGBITS          0000000000001060 00001060
     0000000000000171 0000000000000000 AX 0 0 16
 [17] .fini                PROGBITS          0000000000001104 00001104
     000000000000000d 0000000000000000 AX 0 0 4
 [18] .rodata              PROGBITS          0000000000002000 00002000
     000000000000001c 0000000000000000 A 0 0 4
 [19] .eh_frame_hdr        PROGBITS          000000000000201c 0000201c
     0000000000000044 0000000000000000 A 0 0 4
 [20] .eh_frame            PROGBITS          0000000000002060 00002060
```

```
divo@MSI: ~  
11  
12      .init_array .fini_array .dynamic .got  
divo@MSI:~$ objdump -d program | head -50  
program:      file format elf64-x86-64  
  
Disassembly of section .init:  
0000000000001000 <.init>:  
1000:      f3 0f 1e fa                endbr64  
1004:      48 83 ec 08                sub    $0x8,%rsp  
1008:      48 8b 85 d9 2f 00 00      mov    0x2fd9(%rip),%rax    # 3fe8 <__gmon_start__@Base>  
100f:      48 85 c0                  test   %rax,%rax  
1012:      74 02                     je     1016 <.init+0x16>  
1014:      ff d0                     call   *%rax  
1016:      48 83 c4 08                add    $0x8,%rsp  
101a:      c3                       ret  
  
Disassembly of section .plt:  
0000000000001020 <.plt>:  
1020:      ff 35 9a 2f 00 00      push  0x2f9a(%rip)    # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>  
1026:      ff 25 9c 2f 00 00      jmp    *0x2f9c(%rip)    # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>  
102c:      0f 1f 40 00                nopl   0x0(%rax)  
1030:      f3 0f 1e fa                endbr64  
1034:      68 00 00 00 00          push  $0x0  
1039:      e9 e2 ff ff              jmp     1020 <.init+0x20>  
103e:      66 90                     xchg   %ax,%ax  
  
Disassembly of section .plt.got:  
0000000000001040 <._cxa_finalize@plt>:  
1040:      f3 0f 1e fa                endbr64  
1044:      ff 25 ae 2f 00 00      jmp    *0x2fae(%rip)    # 3ff8 <._cxa_finalize@GLIBC_2.2.5>  
104a:      66 0f 1f 44 00 00      nopw   0x0(%rax,%rax,1)  
  
Disassembly of section .plt.sec:  
0000000000001050 <printf@plt>:  
1050:      f3 0f 1e fa                endbr64  
1054:      ff 25 76 2f 00 00      jmp    *0x2f76(%rip)    # 3fd0 <printf@GLIBC_2.2.5>  
105a:      66 0f 1f 44 00 00      nopw   0x0(%rax,%rax,1)  
  
Disassembly of section .text:  
0000000000001060 <.start>:  
1060:      f3 0f 1e fa                endbr64
```

7. OS Design and Implementation

Praktik 7.1: Kernel Module Sederhana

Buat file hello_module.c:

Makefile:

Compile dan load:

Praktik 7.2 monitoring kernel


```
divo@MSI:~$ cat /var/log/journal/5bdf82febb8842068ff64df56e9fc3bd/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 3218.158178] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3218.290213] Failed to connect to bus: No such file or directory
[ 3218.390809] systemd-journald[52]: Collecting audit messages is disabled.
[ 3218.489028] systemd-journald[52]: Received client request to flush runtime journal.
[ 3218.489152] systemd-journald[52]: File /var/log/journal/5bdf82febb8842068ff64df56e9fc3bd/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 3274.401152] mini_init (223): drop_caches: 1
[ 3859.477026] mini_init (223): drop_caches: 1
[ 6059.079396] mini_init (223): drop_caches: 1
divo@MSI:~$ lsmod
Module                  Size  Used by
tls                     98384  0
intel_rapl_msrm        16384  0
intel_rapl_common      32768  1 intel_rapl_msrm
kvm_amd                 126976  0
kvm                     970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel           16384  2
battery                20480  0
ac                     16384  0
sch_fq_codel           16384  1
configfs               53248  1
autofs4                46056  0
br_netfilter           28672  0
bridge                 282624  1 br_netfilter
stp                    12288  1 bridge
llc                     12288  2 bridge,stp
ip_tables              28672  0
tun                    53248  0
divo@MSI:~$ modinfo ext4
name:                   ext4
filename:               (builtin)
softdep:                pre: crc32c
license:                GPL
file:                   fs/ext4/ext4
description:            Fourth Extended Filesystem
author:                 Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:                  fs-ext4
alias:                  ext3
alias:                  fs-ext3
divo@MSI:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
divo@MSI:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
divo@MSI:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
```

8. Operating-System Structure

Praktik 8.1: Eksplorasi Struktur Layering

Praktik 8.2: Microkernel vs Monolithic

Praktik 8.3: Program - Virtual File System

Buat file procs_read.c:

```
divo@MSI:~$ cat /proc/cpuinfo
processor       : 0
vendor_id     : AuthenticAMD
cpu family    : 25
model         : 68
model name    : AMD Ryzen 5 7535HS with Radeon Graphics
stepping      : 1
microcode    : 0xffffffff
cpu MHz       : 3293.813
cache size   : 512 KB
physical id   : 0
siblings     : 12
core id      : 0
cpu cores    : 6
apicid       : 0
initial apicid : 0
fpu          : yes
fpu_exception : yes
cpuid level  : 13
wp           : yes
flags        : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp la constant_tsc rep_good no
pl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq sse4_3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy
abm sse4a misalignsse 3dnowprefetch osvw topext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bti1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec
xgetbv1 xsaves clzero xsaveerptr arat npt nr1p_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrc
bugs          : sysret_ss_attrs null_seg spectre_v1 spectre_v2 spec_store_bypass srso ibpb_no_ret
bogomips      : 6587.62
tlb size      : 2560 4K pages
clflush size  : 64
cache_alignm  : 64
address sizes : 48 bits physical, 48 bits virtual
power management:

processor      : 1
vendor_id     : AuthenticAMD
cpu family    : 25
model         : 68
model name    : AMD Ryzen 5 7535HS with Radeon Graphics
stepping      : 1
microcode    : 0xffffffff
cpu MHz       : 3293.813
cache size   : 512 KB
physical id   : 0
siblings     : 12
```

```
divo@MSI:~$ cat /etc/os-release
NAME="Linux MSI"
VERSION="6.6.87.2-microsoft-standard-WSL2"
ID="ubuntu"
ID_LIKE="debian"
PRETTY_NAME="Linux MSI 6.6.87.2-microsoft-standard-WSL2"
VERSION_ID="6.6.87.2"
BUILD_ID="20250806.1"
HOME_URL="https://www.ubuntu.com/"
SUPPORT="https://ubuntu.com/forums"

```

```
divo@MSI:~$ uname -a
Linux MSI 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 GNU/Linux
divo@MSI:~$ cat /proc/sys/kernel/ostype
Linux
divo@MSI:~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
divo@MSI:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1   0.0  0.1 22244 12524 ?        Ss   Oct15   0:05 /sbin/init
root           2   0.0  0.0   3072 1792 ?        Sl   Oct15   0:00 /init
root           6   0.0  0.0   3088 1792 ?        Sl   Oct15   0:00 /usr/lib/systemd/systemd --control=socket 7
root          52   0.0  0.2 50444 16128 ?        Ss   Oct15   0:00 /usr/lib/systemd/systemd
root          98   0.0  0.0 25400 6528 ?        Ss   Oct15   0:01 /usr/lib/systemd/systemd
systemd+     168   0.0  0.1 21452 12800 ?        Ss   Oct15   0:00 /usr/lib/systemd/systemd
systemd+     172   0.0  0.1 91020 7680 ?        Ssl  Oct15   0:00 /usr/lib/systemd/systemd
root         178   0.0  0.0  4236 2560 ?        Ss   Oct15   0:00 /usr/sbin/cron -f -p
message+    179   0.0  0.0  9652 5248 ?        Ss   Oct15   0:00 /usr/sbin/cron --system --
root        186   0.0  0.1 18160 8320 ?        Ss   Oct15   0:00 /usr/lib/systemd/systemd
root        193   0.0  0.1 1756096 12672 ?       Ssl  Oct15   0:00 /usr/libexec/utmp-pro-sear
syslog      213   0.0  0.0 222508 5248 ?        Ssl  Oct15   0:00 /usr/sbin/rsyslogd -n -i
root        216   0.0  0.0   3116 1792 tty1    S+   Oct15   0:00 /sbin/agetty -o -p -- \u
root        219   0.0  0.3 106996 22528 ?       Ssl  Oct15   0:00 /usr/bin/python3 /usr/sh
root        295   0.0  0.0   3076 1824 ?        Ss   Oct15   0:00 /init
root        296   0.0  0.0   3092 1152 ?        S   Oct15   0:01 /init
divo        297   0.0  0.0   6204 5248 pts/0    Ss   Oct15   0:00 -bash
root        298   0.0  0.0   6824 4224 pts/1    Ss   Oct15   0:00 /bin/login -f
divo        332   0.0  0.1 20312 11008 ?       Ss   Oct15   0:00 /usr/lib/systemd/systemd
divo        333   0.0  0.0   21148 3520 ?        S   Oct15   0:00 (sd-pam)
divo        360   0.0  0.0   6056 4992 pts/1    S+   Oct15   0:00 -bash
polkitd     813   0.0  0.1 308160 7552 ?        Ssl  Oct15   0:00 /usr/lib/polkit-1/polkitd
root        3200   0.0  0.1 12020 7936 ?        Ss   Oct15   0:00 sshd: /usr/sbin/ssh -D
divo        5326   0.0  0.0   8280 4224 pts/0    R+   00:57   0:00 ps aux

divo@MSI:~$ top
top - 00:58:48 up 3:05, 1 user, load average: 0.02, 0.01, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0/0.0   0%
MiB Mem : 6783.2 total, 5919.4 free, 662.2 used, 368.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 6121.0 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   22244 12524 9196 S   0.2   0.2   0:05.16 systemd
    2 root        20   0   3072 1792 1664 S   0.0   0.0   0:00.00 init-systemd(Ub
    6 root        20   0   3088 1792 1792 S   0.0   0.0   0:00.00 init
   52 root       19  -1 50444 16128 15232 S   0.2   0.2   0:00.76 systemd-journal
   98 root        20   0 25400 6528 4992 S   0.1   0.1   0:01.37 systemd-udevd
  168 systemd+    20   0 21452 12800 10624 S   0.2   0.2   0:00.16 systemd-resolve
  172 systemd+    20   0 91020 7680 6784 S   0.1   0.1   0:00.23 systemd-timesyn
  178 root        20   0  4236 2560 2432 S   0.0   0.0   0:00.02 cron
  179 message+    20   0  9652 5248 4480 S   0.1   0.0   0:00.70 dbus-daemon
  186 root        20   0 18160 8320 7424 S   0.1   0.1   0:00.37 systemd-logind
  193 root        20   0 1756096 12672 16240 S   0.2   0.2   0:00.21 wsl-pro-service
  213 syslog      20   0 222508 5248 4688 S   0.1   0.1   0:00.19 rsyslogd
  216 root        20   0   3116 1792 1664 S   0.0   0.0   0:00.01 agetty
  219 root        20   0 106996 22528 13184 S   0.3   0.3   0:00.11 unattended-upgr
  295 root        20   0   3076 1824 896 S   0.0   0.0   0:00.00 SessionLeader
  296 root        20   0   3092 1152 1824 S   0.0   0.0   0:01.05 Relay(297)
  297 divo         20   0   6204 5248 3584 S   0.1   0.1   0:00.47 bash
  298 root        20   0   6824 4224 3712 S   0.1   0.1   0:00.01 login
  332 divo         20   0 20312 11008 9088 S   0.2   0.2   0:00.15 systemd
  333 divo         20   0 21148 3520 1792 S   0.1   0.1   0:00.00 (sd-pam)
  360 divo         20   0   6056 4992 3456 S   0.1   0.1   0:00.03 bash
  813 polkitd     20   0 308160 7552 6784 S   0.1   0.1   0:00.05 polkitd
 3200 root        20   0 12020 7936 6912 S   0.1   0.1   0:00.00 sshd
 5327 divo         20   0  9316 5632 3456 R   0.1   0.1   0:00.03 top
```

```
divo@MSI:~$ top
top - 00:58:48 up 3:05, 1 user, load average: 0.02, 0.01, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0/0.0   0%
MiB Mem : 6783.2 total, 5919.4 free, 662.2 used, 368.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 6121.0 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   22244 12524 9196 S   0.2   0.2   0:05.16 systemd
    2 root        20   0   3072 1792 1664 S   0.0   0.0   0:00.00 init-systemd(Ub
    6 root        20   0   3088 1792 1792 S   0.0   0.0   0:00.00 init
   52 root       19  -1 50444 16128 15232 S   0.2   0.2   0:00.76 systemd-journal
   98 root        20   0 25400 6528 4992 S   0.1   0.1   0:01.37 systemd-udevd
  168 systemd+    20   0 21452 12800 10624 S   0.2   0.2   0:00.16 systemd-resolve
  172 systemd+    20   0 91020 7680 6784 S   0.1   0.1   0:00.23 systemd-timesyn
  178 root        20   0  4236 2560 2432 S   0.0   0.0   0:00.02 cron
  179 message+    20   0  9652 5248 4480 S   0.1   0.0   0:00.70 dbus-daemon
  186 root        20   0 18160 8320 7424 S   0.1   0.1   0:00.37 systemd-logind
  193 root        20   0 1756096 12672 16240 S   0.2   0.2   0:00.21 wsl-pro-service
  213 syslog      20   0 222508 5248 4688 S   0.1   0.1   0:00.19 rsyslogd
  216 root        20   0   3116 1792 1664 S   0.0   0.0   0:00.01 agetty
  219 root        20   0 106996 22528 13184 S   0.3   0.3   0:00.11 unattended-upgr
  295 root        20   0   3076 1824 896 S   0.0   0.0   0:00.00 SessionLeader
  296 root        20   0   3092 1152 1824 S   0.0   0.0   0:01.05 Relay(297)
  297 divo         20   0   6204 5248 3584 S   0.1   0.1   0:00.47 bash
  298 root        20   0   6824 4224 3712 S   0.1   0.1   0:00.01 login
  332 divo         20   0 20312 11008 9088 S   0.2   0.2   0:00.15 systemd
  333 divo         20   0 21148 3520 1792 S   0.1   0.1   0:00.00 (sd-pam)
  360 divo         20   0   6056 4992 3456 S   0.1   0.1   0:00.03 bash
  813 polkitd     20   0 308160 7552 6784 S   0.1   0.1   0:00.05 polkitd
 3200 root        20   0 12020 7936 6912 S   0.1   0.1   0:00.00 sshd
 5327 divo         20   0  9316 5632 3456 R   0.1   0.1   0:00.03 top
```

```
divo@MSI:~$ which bash
/usr/bin/bash
divo@MSI:~$ which gcc
/usr/bin/gcc
divo@MSI:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
divo@MSI:~$ cat /boot/config-$(uname -r) | grep "zy" | wc -l # Built-in
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
divo@MSI:~$ cat /boot/config-$(uname -r) | grep "m" | wc -l # Modules
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
divo@MSI:~$ lsmod | wc -l
10
divo@MSI:~$ nano procfs_read.c
divo@MSI:~$ ^C
divo@MSI:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:32:60: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   32 |     snprintf(proc_path, sizeof(proc_path), "/proc/kd/status", getpid());
divo@MSI:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:32:60: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   32 |     snprintf(proc_path, sizeof(proc_path), "/proc/kd/status", getpid());
divo@MSI:~$ dmesg | less
divo@MSI:~$ systemd-analyze
Startup finished in 1.633s (userspace)
graphical.target reached after 1.588s in userspace.
divo@MSI:~$ systemd-analyze blame
9.781s motd-news.service
636ms landscape-client.service
407ms dev-sda.device
296ms snapd.seeded.service
291ms man-db.service
248ms systemd-udev-trigger.service
239ms console-getty.service
210ms user@1000.service
207ms my_daemon.service
167ms wsl-pro.service
126ms snapd.service
124ms systemd-resolved.service
96ms rsyslog.service
91ms systemd-journal-flush.service
89ms systemd-journald.service
88ms systemd-timesyncd.service
82ms keyboard-setup.service
77ms dpkg-db-backup.service
69ms systemd-logind.service
65ms systemd-logind.service
```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

Praktik 9.2: Kernel Parameters

Praktik 9.3: Simple Bootloader (Konsep)

```
divo@MSI:~$ dmesg | less
divo@MSI:~$ systemd-analyze
Startup finished in 1.633s (userspace)
graphical.target reached after 1.588s in userspace.
divo@MSI:~$ systemd-analyze blame
9.781s motd-news.service
636ms landscape-client.service
407ms dev-sda.device
296ms snapd.seeded.service
291ms man-db.service
248ms systemd-udev-trigger.service
239ms console-getty.service
210ms user@1000.service
207ms my_daemon.service
167ms wsl-pro.service
126ms snapd.service
124ms systemd-resolved.service
96ms rsyslog.service
91ms systemd-journal-flush.service
89ms systemd-journald.service
88ms systemd-timesyncd.service
82ms keyboard-setup.service
77ms dpkg-db-backup.service
69ms systemd-logind.service
65ms systemd-tmpfiles-setup-dev.service
60ms systemd-udev.service
51ms systemd-tmpfiles-setup.service
51ms user-runtime-dir@1000.service
47ms polkit.service
44ms e2scrub_reap.service
42ms logrotate.service
40ms dev-hugepages.mount
39ms dev-mqueue.mount
38ms sys-kernel-debug.mount
37ms sys-kernel-tracing.mount
33ms dbus.service
30ms kmod-static-nodes.service
29ms modprobe@configfs.service
28ms modprobe@dm_mod.service
28ms modprobe@drm.service
27ms modprobe@efi_pstore.service
26ms modprobe@fuse.service
25ms ssh.service
25ms modprobe@loop.service
25ms systemd-tmpfiles-clean.service
23ms systemd-tmpfiles-setup-dev-early.service
23ms snapd.socket
22ms systemd-modules-load.service
20ms systemd-remount-fs.service
19ms systemd-update-utmp.service
19ms systemd-user-sessions.service
19ms systemd-sysctl.service
```

```
divo@MSI: ~  
14ms sys-kernel-config.mount  
12ms setvtrgb.service  
12ms systemd-update-utmp-runlevel.service  
divo@MSI:~$ systemd-analyze critical-chain  
The time when unit became active or started is printed after the "@" character.  
The time the unit took to start is printed after the "+" character.  
  
graphical.target @1.588s  
└─multi-user.target @1.588s  
   └─snapd.seeded.service @954ms +296ms  
      └─basic.target @937ms  
         └─sockets.target @937ms  
            └─snapd.socket @913ms +23ms  
               └─systemd-init.target @904ms  
                  └─systemd-resolved.service @778ms +124ms  
                     └─systemd-tmpfiles-setup.service @530ms +54ms  
                        └─systemd-journal-flush.service @435ms +91ms  
                           └─systemd-journald.service @322ms +89ms  
                              └─systemd-journald.socket @287ms  
                                 └─system.slice @276ms  
                                    └─.slice @276ms  
                                       └─.slice @276ms  
divo@MSI:~$ cat /boot/grub/grub.cfg | head -50  
cat: /boot/grub/grub.cfg: No such file or directory  
divo@MSI:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20  
Command 'lsinitramfs' not found, but can be installed with:  
sudo apt install initramfs-tools-core  
divo@MSI:~$ cat /proc/cmdline  
initrd=initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1  
divo@MSI:~$ sudo nano /etc/default/grub  
divo@MSI:~$ nano bootloader-concept.asm  
divo@MSI:~$ strace ls  
Command 'strace' not found, but can be installed with:  
sudo apt install strace  
divo@MSI:~$ strace -v ls  
Command 'strace' not found, but can be installed with:  
sudo apt install strace  
divo@MSI:~$ strace -c ls  
Command 'strace' not found, but can be installed with:  
sudo apt install strace  
divo@MSI:~$ strace -o trace.txt ls  
Command 'strace' not found, but can be installed with:  
sudo apt install strace  
divo@MSI:~$ cat trace.txt  
cat: trace.txt: No such file or directory  
divo@MSI:~$ ltrace ls  
Command 'ltrace' not found, but can be installed with:  
sudo apt install ltrace  
divo@MSI:~$ ltrace -c ls  
Command 'ltrace' not found, but can be installed with:  
sudo apt install ltrace  
divo@MSI:~$ sudi apt install ltrace  
Command 'sudi' not found, did you mean:  
command 'sudo' from deb sudo (1.9.15p5-3ubuntu5.24.04.1)
```

10. Operating-System Debugging

Praktik 10.1: Using strace

Praktik 10.2: Using ltrace

Praktik 10.3: Using GDB

Praktik 10.4: Kernel Debugging dengan dmesg

Praktik 10.5: Memory Leak Detection dengan Valgrind

Praktik 10.6: Performance Analysis

```
divo@MSI:~$ strace -o trace.txt ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
divo@MSI:~$ cat trace.txt
cat: trace.txt: No such file or directory
divo@MSI:~$ ltrace ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
divo@MSI:~$ ltrace -c ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
divo@MSI:~$ sudi apt install ltrace
Command 'sudi' not found, did you mean:
  command 'sudo' from deb sudo (1.9.15p5-3ubuntu5.24.04.1)
  command 'sudo' from deb sudo-ldap (1.9.15p5-3ubuntu5.24.04.1)
Try: sudo apt install -deb name>
divo@MSI:~$ sudo apt install ltrace
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ltrace
0 upgraded, 1 newly installed, 0 to remove and 188 not upgraded.
Need to get 134 kB of archives.
After this operation, 391 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 ltrace amd64 0.7.3-6.4ubuntu3 [134 kB]
Fetched 134 kB in 2s (88.9 kB/s)
Selecting previously unselected package ltrace.
(Reading database ... 48827 files and directories currently installed.)
Preparing to unpack .../ltrace.0.7.3-6.4ubuntu3_amd64.deb ...
Unpacking ltrace (0.7.3-6.4ubuntu3) ...
Setting up ltrace (0.7.3-6.4ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
divo@MSI:~$ nano buggy.c
divo@MSI:~$ gcc -g -o buggy buggy.c
divo@MSI:~$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
divo@MSI:~$ (gdb) break main
-Bash: syntax error near unexpected token `break'
divo@MSI:~$ sudo apt install gdb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libdw1t64 libelf1t64
  libipt2 libsource-highlight-common libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2
  libsource-highlight-common libsource-highlight4t64
The following packages will be upgraded:
  libdw1t64 libelf1t64
2 upgraded, 8 newly installed, 0 to remove and 186 not upgraded.
Need to get 12.3 MB of archives.
After this operation, 26.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu0.1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdw1t64 amd64 0.190-1.1ubuntu0.1 [261 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libelf1t64 amd64 0.190-1.1ubuntu0.1 [57.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.5.11-3build3 [164 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod1t64 amd64 0.190-1.1ubuntu0.1 [17.1 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.0.6-1build1 [45.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-common all 3.1.9-4.3build1 [64.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build1 [258 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 gdb amd64 15.0.50.20240803-0ubuntu1 [4019 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-dbg amd64 2.39-0ubuntu6.6 [7458 kB]
Fetched 12.3 MB in 4s (2908 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 48837 files and directories currently installed.)
Preparing to unpack .../0-libdebuginfod-common.0.190-1.1ubuntu0.1_all.deb ...
Unpacking libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Preparing to unpack .../1-libdw1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdw1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Preparing to unpack .../2-libelf1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libelf1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../5-libbabeltrace1.1.5.11-3build3_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-3build3) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../4-libdebuginfod1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.190-1.1ubuntu0.1) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../5-libipt2.2.0.6-1build1_amd64.deb ...
Unpacking libipt2 (2.0.6-1build1) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../6-libsource-highlight-common.3.1.9-4.3build1_all.deb ...
```

```
divo@MSI:~$ sudo apt install gdb
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
divo@MSI:~$ (gdb) break main
-Bash: syntax error near unexpected token `break'
divo@MSI:~$ sudo apt install gdb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libdw1t64 libelf1t64
  libipt2 libsource-highlight-common libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2
  libsource-highlight-common libsource-highlight4t64
The following packages will be upgraded:
  libdw1t64 libelf1t64
2 upgraded, 8 newly installed, 0 to remove and 186 not upgraded.
Need to get 12.3 MB of archives.
After this operation, 26.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu0.1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdw1t64 amd64 0.190-1.1ubuntu0.1 [261 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libelf1t64 amd64 0.190-1.1ubuntu0.1 [57.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.5.11-3build3 [164 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod1t64 amd64 0.190-1.1ubuntu0.1 [17.1 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.0.6-1build1 [45.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-common all 3.1.9-4.3build1 [64.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build1 [258 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 gdb amd64 15.0.50.20240803-0ubuntu1 [4019 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-dbg amd64 2.39-0ubuntu6.6 [7458 kB]
Fetched 12.3 MB in 4s (2908 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 48837 files and directories currently installed.)
Preparing to unpack .../0-libdebuginfod-common.0.190-1.1ubuntu0.1_all.deb ...
Unpacking libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Preparing to unpack .../1-libdw1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdw1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Preparing to unpack .../2-libelf1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libelf1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../5-libbabeltrace1.1.5.11-3build3_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-3build3) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../4-libdebuginfod1t64.0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.190-1.1ubuntu0.1) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../5-libipt2.2.0.6-1build1_amd64.deb ...
Unpacking libipt2 (2.0.6-1build1) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../6-libsource-highlight-common.3.1.9-4.3build1_all.deb ...
```